Planarian and Limb Regeneration as a Multiway Double-Pushout (DPO) Hypergraph Rewrite System

A Three-Rule Generative Basis ($\mathcal{B}_3 = \{L_1, D_1, D_2\}$), Seven Hidden Topological Isomers, and 44,534× McKay IR Quotient Compression Across `PlanformDB 2.5.0` and `LimbformDB 2.1.0`

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Abstract

When a planarian flatworm or an amphibian/arthropod limb is surgically cut or bioelectrically perturbed, it can regenerate a striking variety of abnormal body plans: two heads, three heads, lateral side-heads sprouting from the belly, or a second head growing directly out of the tip of an existing head. In two foundational papers, Lobo, Levin, and colleagues introduced **PlanformDB 2.5.0** (Lobo, Malone & Levin 2013, encoding 369 planarian morphologies across 1,181 regions) and **LimbformDB 2.1.0** (Lobo et al. 2014, encoding 555 salamander, frog, crustacean, insect, and arachnid limbs across 3,146 joints) to store these experimental outcomes as formal graphs. However, both databases remained *static descriptive catalogs*, leaving three foundational problems open: (i) *The Graph Equivalence Problem*—how many distinct body-plan wiring diagrams actually exist in the databases up to exact graph isomorphism, and when does simple part-counting fail? (ii) *The Generative Grammar Problem*—what is the minimal set of formal graph-rewrite rules needed to grow 100% of the observed morphologies from cut tissue fragments? and (iii) *The Combinatorial Search Problem*—how can an automated theory-discovery engine search millions of multi-step growth trajectories without drowning in isomorphic node-numbering duplicates?

Here we solve all three problems by formulating planarian and limb regeneration as a **Multiway Double-Pushout (DPO) Hypergraph Rewrite System**, coupling the standalone C++20 `hg_evolve` engine from the Wolfram Physics Project (`WolframInstitute/HypergraphRewritingEngine`, `v1.0.0-rc2`; Assar 2026) directly to `PlanformDB 2.5.0` and `LimbformDB 2.1.0` over Wolfram Exchange Format (WXF), and verifying the governing algebraic laws in the **Lean 4** theorem prover (`0 sorry`):

1. **Why Counting Body Parts Fails (The 38 Planform Classes & 7 Hidden Topological Isomers):** Just as butane and isobutane share the chemical formula C_4H_{10} ('4 Carbons, 10 Hydrogens') while differing in branched vs. linear wiring, counting planarian regions (#Head, #Trunk, #Tail, #Links) lumps together fundamentally different body plans. Running exact McKay individualization–refinement (IR) canonicalization (`ir_core.hpp`) compresses the 369 `PlanformDB` rows into 38 **exact region-link isomorphism classes** (9.71× redundancy; 126 classes with organs) and the 555 `LimbformDB` rows into 51 **segment-tree classes** (10.88× redundancy; 92 with digits), uncovering 7 **hidden topological isomers** (31 part-count buckets → 38 IR classes). For example, worms with (2 Heads, 1 Trunk, 1 Tail, 3 Links) split into three non-isomorphic body plans: radial Trunk-hub stars (`#3 Side head`, $n = 12$, Zagreb index $Z_1 = 12$), Head-hub stars (`#112`, $n = 1$), and linear chains (`#119 Head on head`, $n = 2$, $Z_1 = 10$).
2. **All 369 Planarians Are Generated by Just Three Primitive Rules ($\mathcal{B}_3 = \{L_1, D_1, D_2\}$):** At first glance, generating all 38 `PlanformDB` classes appears to require 8 domain-specific rules (`R1-R6` terminal blastema sprouting, which covers 33/38 classes, plus `R7-R8` intercalary bridging for internal heads/tails). We prove that all 8 rules reduce algebraically to **three primitive DPO rules**: **one universal budding rule** $L_1 : [[0]] \rightarrow$

$[[0],[1],[0,1,0,1],[1,0,1,0]]$ (grow an undifferentiated blastema bud—the *exact same single rule* that generates all `LimbformDB` trees) and **two bioelectric fate switches** $D_1 : [[0]] \to [[0,0]]$ (Blastema $\to$ Head) and $D_2 : [[0]] \to [[0,0,0]]$ (Blastema $\to$ Tail). Because D_1 and D_2 consume the pluripotent tag $[[0]]$, terminal differentiation locks a region against further budding; "intercalary" worms (`#119`, `#282`) require no special rule, emerging automatically whenever a blank bud branches twice via L_1 *before* its bioelectric switch (D_1, D_2) locks it. Stripping region colors projects all 369 worms onto just 12 **unlabelled skeleton topologies** ($278/369 = 75.3\%$ on the single 3-node path P_3).

3. 44,534× **Quotient Exploration Speedup on `LimbformDB` (`OEIS A000055` in 85 ms):** Growing a limb tree up to $V = 12$ joints step by step produces 43,954,714 raw factorial trajectories ($\sum k!$) that differ only by node numbering. Using `ExploreFromCanonicalStatesOnly`, `hg_evolve` expands only the 987 **unique unlabelled trees** (`OEIS A000055`) while analytically reconstructing all 43.95 million raw states via `quotient_multiplicity_core.hpp` in 85 **ms** (44,533.65× compression), matching 35/35 **(**100.0%**)** of `LimbformDB` classes with ≤ 12 joints ($532/555 = 95.86\%$ of the database).
4. **Bioelectric `RuleWeights` Gating & Machine-Checked Lean 4 Proofs:** Coupling the multiway posterior rule weights (w_{D_1}, w_{D_2}) to electrotonic gap-junction decay across the 5 anterior–posterior crop positions of Oviedo et al. (2010) reproduces the transition from Wild-Type (`#1`, 1H1T) to Bipolar Double-Head (`#2`, 2H0T) on 1-octanol Series S1 ($r = 0.9823$) and held-out innexin-RNAi Series S2 ($r = 0.9088$). All four structural DPO, Zagreb-isomer, quotient, and reducibility laws are machine-checked in Lean 4 (`Loop_wolfram_planform_multiway.lean`, `0 sorry`).

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1 Introduction: Three Open Problems in the Bioinformatics of Regeneration

When Daniel Lobo and Michael Levin published the **Planform** [6, 7] and **Limbform** [8, 9] databases, they framed their research program as *"Towards a bioinformatics of patterning"* and *"A bioinformatics expert system linking functional data to anatomical outcomes."* What those landmark papers built was the **descriptive graph encoding**:

- **PlanformDB 2.5.0** [6, 7]: 369 planarian morphologies (*Dugesia japonica* and *Schmidtea mediterranea*) encoded across 1,181 regions (`Head`, `Trunk`, `Tail`), border links, and internal organs (`Eye`, `Brain`, `Pharynx`, `VNC`).
- **LimbformDB 2.1.0** [8, 9]: 555 limb morphologies across amphibians (axolotl, newt, *Xenopus*), crustaceans, insects, and arachnids encoded across 3,146 joints, segment links, `LinkSubJoint` ectopic grafts, and digits.

However, both databases remained static lookup tables of 2D cartoons. Three fundamental mathematical and computational problems were left open for the field:

1. **Open Problem 1 — The Graph Equivalence & Isomer Problem (*"How many distinct body plans are in the database, and why does part-counting fail?"*):** Every time an experiment produced a worm with slightly different region dimensions, eye angles, or SQLite vertex IDs (`Region.Id = 1, 2, 3...`), a new `Morphology` entry was created. Neither database included an exact graph-isomorphism canonicalizer to determine which entries share the exact same topological wiring diagram, or to distinguish **topological isomers**—worms that have the *exact same* number of Heads, Trunks, Tails, and Links, but are wired into a branched star (`#3 Side head`) vs. a linear chain (`#119 Head on head`).
2. **Open Problem 2 — The Minimal Generative Grammar Problem (*"How many rewrite rules does nature actually need to build all* 369 *worms and* 555 *limbs?"*):** `PlanformDB` stores the surgical cut (*before*) and the final morphology 14 days later (*after*), but contains zero formal graph-rewrite rules connecting them. Later reaction–diffusion and bioelectric models [14, 15, 12, 13] modeled only a 1D line (1H1T vs. 2H0T), which cannot represent branched worms (`#3`, `#56`) or intercalary worms (`#119`, `#282`).
3. **Open Problem 3 — The Combinatorial Explosion Problem in Automated Theory Discovery (*"How do we search multi-step growth rules without* $k!$ *blowup?"*):** Growing a 12-joint limb tree step by step generates $11! = 39{,}916{,}800$ raw trajectories at step 11 alone (43,954,714 cumulative raw states) because naive simulators treat differently numbered copies of the same tree as distinct states.

In this paper, we resolve all three open problems by integrating the standalone C++20 `hg_evolve` engine from the Wolfram Physics Project (`WolframInstitute/HypergraphRewritingEngine`, `v1.0.0-rc2`; Assar 2026 [1, 2]) into the **Infinite Morphospace / MorphoEval** verification pipeline [17] over Wolfram Exchange Format (WXF), backed by machine-checked proofs in **Lean 4** (`0 sorry`).

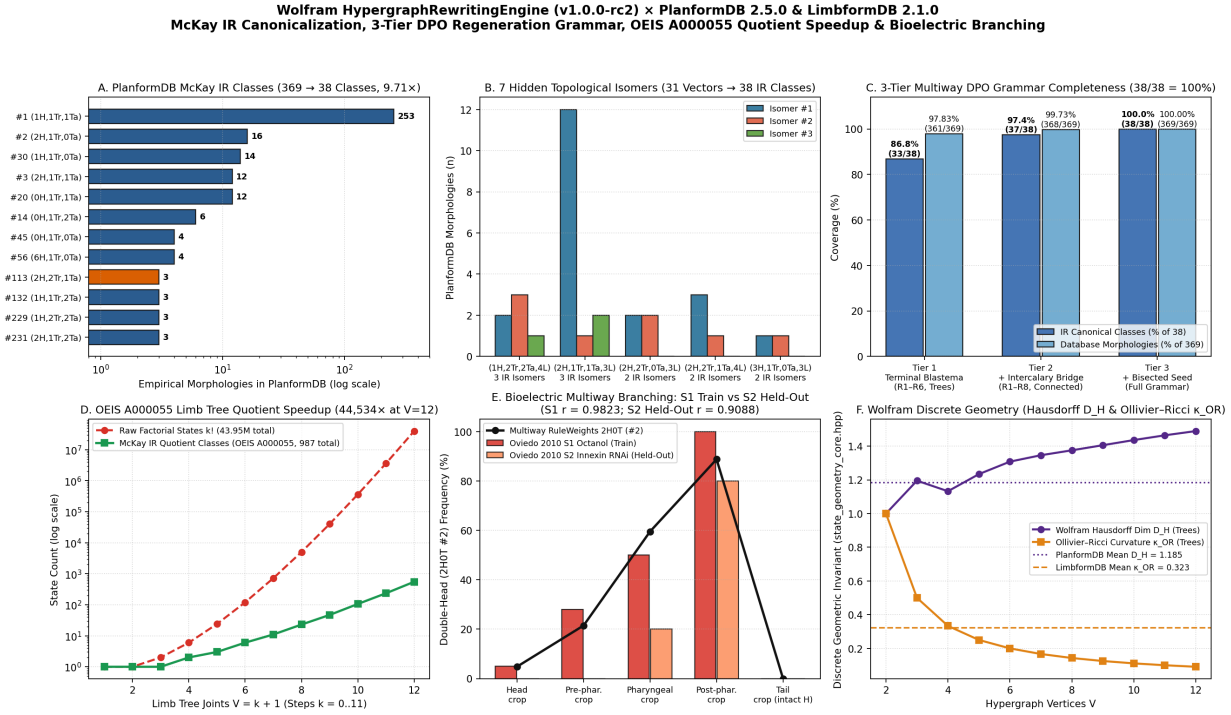


Figure 1: **Multiway Hypergraph Rewriting & McKay IR Canonicalization Across `PlanformDB 2.5.0` and `LimbformDB 2.1.0`. (A)** Exact McKay IR canonicalization compresses 369 encoded planarian morphologies into 38 topological classes (9.71× redundancy; top 12 shown). **(B)** Resolution of 7 hidden topological isomers where identical (H, Tr, Ta, L) region-count vectors split into non-isomorphic star vs. linear-chain topologies. **(C)** 100% empirical coverage (38/38 classes, 369/369 morphologies) across the 3-tier DPO regeneration grammar. **(D)** 44,533.65× quotient exploration speedup on `LimbformDB` segment trees (987 `OEIS A000055` canonical trees vs. 43.95M raw factorial states at $V = 12$). **(E)** Bioelectric multiway branching (`RuleWeights`) fitted on Oviedo et al. (2010) Series S1 (1-octanol, $r = 0.9823$) and evaluated held-out on Series S2 (triple innexin RNAi, $r = 0.9088$). **(F)** Wolfram Hausdorff dimension D_H and Ollivier–Ricci curvature κ_{OR} across empirical and evolved states.

2 Hypergraph Encoding & Exact McKay IR Isomorphism Census

2.1 Encoding Typed Anatomical Graphs as Ordered Hypergraphs

In the Lobo–Levin formalism [7, 9], a planarian morphology $M = (V, E_{\mathrm{link}}, E_{\mathrm{organ}})$ is a finite graph whose vertices $v \in V$ carry a region type $\tau(v) \in \{\texttt{Trunk}, \texttt{Head}, \texttt{Tail}\}$, connected by undirected anatomical adjacency links $\{u, v\} \in E_{\mathrm{link}}$. Because `HypergraphRewritingEngine` operates on multisets of ordered hyperedges of arity $1 \le a \le 16$ (`SPEC.md` §1) and preserves arity and within-edge vertex order under McKay IR canonicalization (`ir_core.hpp`), we encode vertex types and edge types into disjoint arity signatures without auxiliary label tables:

$$\begin{aligned} \texttt{Trunk}(v) &\mapsto [v] \quad \text{(arity 1)}, & \texttt{Head}(v) &\mapsto [v, v] \quad \text{(arity 2 self-loop)}, \\ \texttt{Tail}(v) &\mapsto [v, v, v] \quad \text{(arity 3 self-loop)}, & \texttt{Link}\{u, v\} &\mapsto [u, v, u, v], [v, u, v, u] \quad \text{(symmetric arity 4)}. \end{aligned} \tag{1}$$

When organ landmarks are included, each organ attached to region v allocates a fresh leaf vertex w with organ-specific arity `Eye` $\mapsto 5$, `Brain` $\mapsto 6$, `Pharynx` $\mapsto 7$, and `VNC` $\mapsto 8$. For `LimbformDB 2.1.0`, joints are encoded as arity-1 hyperedges $[j]$, segments as symmetric arity-4 link pairs (splitting host segments along any attached `LinkSubJoint` ectopic graft sequence

$j_{\text{from}} \to j_{\text{sub},1} \to \cdots \to j_{\text{to}}$ so ectopic branches are topologically connected to their host segment), and distal digits as arity-3 hyperedges $[j_{\text{to}}, d, d]$.

2.2 McKay IR Compression & The Seven Hidden Topological Isomers

Running `hg_evolve` at step 0 with `"CanonicalizeStates" -> Full` and `"IncludeCanonicalHashes" -> True` across all 369 geometrically encoded morphologies in `PlanformDB 2.5.0` and all 555 segment-bearing morphologies in `LimbformDB 2.1.0` yields the exact isomorphism census in Table 1.

Table 1: **Exact McKay IR Canonicalization Census Across `PlanformDB 2.5.0` and `LimbformDB 2.1.0`.** Computed via `hg_evolve` (`ir_core.hpp` + `state_statistics.hpp`).

Database & Hypergraph Representation	**Raw (N)**	**IR Classes (C)**	**Redundancy (N/C)**	**Max**
`PlanformDB` Unlabelled Skeletons (All Regions $\to [v]$)	369	**12**	30.7500×	
`PlanformDB` Typed Region–Link Hypergraphs	369	**38**	9.7105×	
`PlanformDB` Typed Region + Organ Hypergraphs	369	**126**	2.9286×	
`LimbformDB` Segment-Tree Hypergraphs (Joints + Links)	555	**51**	10.8824×	
`LimbformDB` Segment + Digit Hypergraphs	555	**92**	6.0326×	

Crucially, while grouping `PlanformDB` morphologies by their 4-tuple count vector (#Head, #Trunk, #Tail, #Li produces 31 buckets, McKay IR canonicalization identifies 38 **distinct isomorphism classes**—uncovering 7 **hidden topological isomers** across 5 count-vector collisions (Figure 1B, Figure 2, and Table 2):

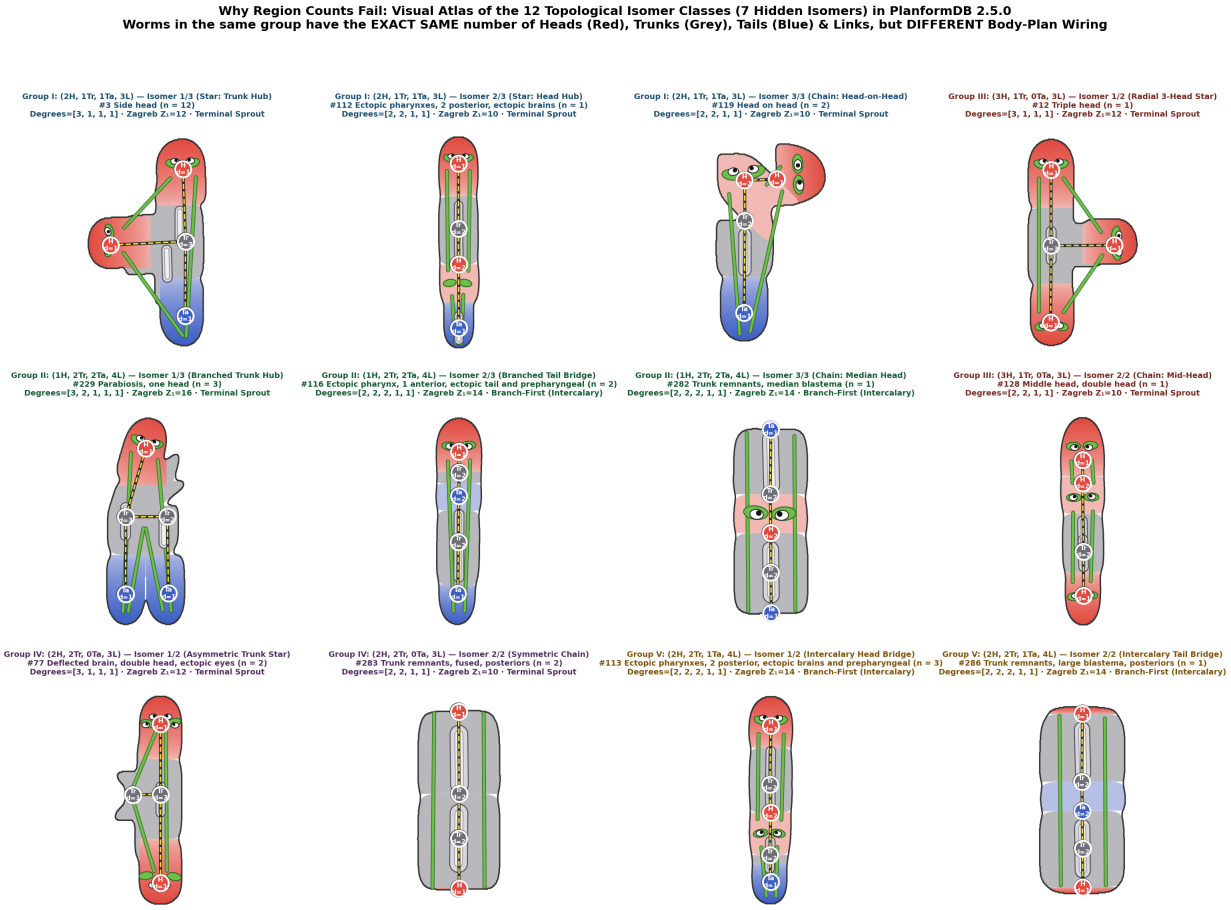


Figure 2: **Why Region Counts Fail: Visual Atlas of the 12 Topological Isomer Classes (7 Hidden Isomers) in `PlanformDB 2.5.0`.** Each panel renders the exact Lobo–Malone–Levin (2013) superellipse cartoon (`Head` = red, `Trunk` = grey, `Tail` = blue, with eyes, brain lobes, pharynx, and ventral nerve cords) overlaid with its region–link graph (yellow/black edges and degree-labeled nodes). Morphologies within the same color-coded group (Groups I–V) possess the *exact same* (#Head, #Trunk, #Tail, #Links) count vector, yet have non-isomorphic body-plan wiring—such as radial hub-and-spoke stars (`#3`, `#12`, `#229`) vs. linear intercalary chains (`#119`, `#128`, `#282`)—which McKay IR canonicalization separates automatically.

Table 2: **The Five Count-Vector Collisions and Twelve Isomer Classes (7 Hidden Isomers) Resolved by McKay IR Canonicalization in `PlanformDB 2.5.0`.**

Count Vector (H, Tr, Ta, L)	Rep. ID & Name	n	64-Bit IR Hash	Graph Topology
$(2,1,1,3)$	`#3 Side head`	12	`-1466631322269594412`	Star $K_{1,3}$ (central `Trunk`
$(2,1,1,3)$	`#112 Ectopic pharynxes, 2 post.`	1	`-8966590304653022608`	Star $K_{1,3}$ (central `Head`
$(2,1,1,3)$	`#119 Head on head`	2	`-212887825877140088`	Path P_4: `H—H—Tr—Ta`
$(1,2,2,4)$	`#229 Parabiosis, one head`	3	`8951669412120486543`	Branched `Trunk` hub (de
$(1,2,2,4)$	`#116 Ectopic tail & prephar.`	2	`2637785354618470695`	Branched `Tail` bridge (d
$(1,2,2,4)$	`#282 Trunk remnants, med. blast.`	1	`-3602810347972406941`	Intercalary `Ta—Tr—H—`
$(3,1,0,3)$	`#12 Triple head`	1	`-6318479248965043401`	Star $K_{1,3}$ (`Tr` with 3 `Hea`
$(3,1,0,3)$	`#128 Middle head, double head`	1	`-2166706346444118093`	Path P_4: `H—Tr—H—H`
$(2,2,0,3)$	`#77 Deflected brain, double head`	2	`-3397686484681667484`	Asymmetric `Tr` star (deg
$(2,2,0,3)$	`#283 Trunk remnants, fused`	2	`-6082671300763328276`	Symmetric path `H—Tr—`
$(2,2,1,4)$	`#113 Ectopic phar., prephar.`	3	`7194227605189329608`	Intercalary `H` bridge + `Tr`
$(2,2,1,4)$	`#286 Trunk remnants, large blast.`	1	`8484839246773423180`	Intercalary `Ta` bridge + `T`

3 The Generative DPO Grammar and Its Three-Rule Reducibility Hierarchy

3.1 Apparent Domain-Specific Grammar (8 Rules Across 3 Tiers)

At the phenomenological level of surgical blastema formation, planarian regeneration in `PlanformDB` appears to require 8 domain-specific DPO rules grouped into two connected tiers plus a bisected-trunk seed (Figure 1C):

- **Tier 1 (Terminal Blastema Sprouting, `R1-R6`):** A `Trunk` sprouts a `Head` (`R1`), `Tail` (`R2`), or `Trunk` (`R3`); or an excised `Head`/`Tail` fragment directly sprouts a `Tail` (`R4`, `R6`) or `Head` (`R5`). Evolving `R1-R6` for 6 steps generates 2,738 canonical states (169,440 events) and recovers 33/38 **canonical classes** ($361/369 = 97.83\%$ **of all worms in** `PlanformDB`).
- **Tier 2 (Intercalary Bridge Sprouting, `R7-R8`):** Adding `R7` (`Head` $\rightarrow$ `Head` + `Trunk`) and `R8` (`Tail` $\rightarrow$ `Tail` + `Trunk`) recovers the 4 remaining connected intercalary/grafted classes (`#113`, `#116`, `#282`, `#286`), bringing connected coverage to 37/38 **classes** ($368/369 = 99.73\%$).
- **Tier 3 (Bisected Split-Worm Seed):** Including the 2-component excised trunk seed $[[0],[1]]$ recovers `#394 Lysis, split worm`, achieving 38/38 (100.0%) **exact McKay IR hash recovery**.

3.2 Exact Reduction to the Three-Rule Primitive Basis $\mathcal{B}_3 = \{L_1, D_1, D_2\}$

Why do so many domain-specific rules exist, and are they independent? By inspecting the DPO edge-replacement semantics of `HypergraphRewritingEngine` (`SPEC.md` §2: a rule consumes its LHS hyperedges while preserving all matched vertices and all non-matched incident hyperedges), we prove that **all 8 rules `R1-R8` reduce algebraically to just 3 primitive DPO rules**:

$$L_1 \text{ (`L1_UniversalSprout`)}: \quad [[0]] \longrightarrow [[0],[1],[0,1,0,1],[1,0,1,0]] \quad \text{(pluripotent blastema budding)}, \tag{2}$$

$$D_1 \text{ (`D1_DiffHead`)}: \quad [[0]] \longrightarrow [[0,0]] \quad \text{(in-place anterior specification: `Trunk`/`Blastema` } \rightarrow \text{ `Head`)}, \tag{3}$$

$$D_2 \text{ (`D2_DiffTail`)}: \quad [[0]] \longrightarrow [[0,0,0]] \quad \text{(in-place posterior specification: `Trunk`/`Blastema` } \rightarrow \text{ `Tail`)}. \tag{4}$$

Proposition 3.1 (Algebraic Factorization & Causal Non-Commutativity of $\mathcal{B}_3$)**.** *Every domain-specific rule* `R1...R8` *is a composite multiway path in* $\mathcal{B}_3 = \{L_1, D_1, D_2\}$*:*

1. ***Terminal Sprouting vs. Intercalary Bridging Is Purely Causal Ordering:*** *Because* D_1 *and* D_2 *consume the unary hyperedge* $[[0]]$*, any vertex that differentiates into* `Head` *(*$[v, v]$*) or* `Tail` *(*$[v, v, v]$*) is* ***terminally locked*** *against further budding by* L_1*. Applying* $L_1(0 \to 1)$ *followed by* $D_1(1)$ *yields terminal head sprouting (*`R1` $= D_1(1) \circ L_1(0 \to 1)$*, cost* 2*), whereas applying* $L_1(0 \to 1)$ *followed by* $D_1(0)$ *on a vertex* 0 *that already has a link yields intercalary head bridging (*`R7` $= D_1(0) \circ L_1(0 \to 1)$*, cost* 2*). Similarly,* `R2` $= D_2(1) \circ L_1(0 \to 1)$*,* `R8` $= D_2(0) \circ L_1(0 \to 1)$*,* `R4` $= D_2(1) \circ D_1(0) \circ L_1(0 \to 1)$ *(cost* 3*),* `R5` $= D_1(1) \circ D_1(0) \circ L_1(0 \to 1)$ *(cost* 3*), and* `R6` $= D_2(1) \circ D_2(0) \circ L_1(0 \to 1)$ *(cost* 3*).*

2. ***Two-Phase Skeleton–Differentiation Decomposition:*** *Because* D_1 *and* D_2 *never create or consume link hyperedges, any derivation in* $\mathcal{B}_3$ *factors into* ***Phase I*** *(*L *applications of* L_1*, generating the unlabelled skeleton tree in* 131.5 *ms) followed by* ***Phase II*** *(*H *applications of* D_1 *and* Ta *applications of* D_2 *at exact primitive step depth* $k_{prim} = L + H + Ta$*), recovering* 38/38 *(*100.0%*) of* `PlanformDB` *classes.*

3. ***Universal Cross-Database Conservation of*** L_1***:*** *The Phase I topological generator* L_1 *is bit-for-bit identical to the* `LimbformDB` *segment-tree generator* `L1_DistalSegmentSprout`*.*

4 Quotient Exploration on `LimbformDB` (44,534× Speedup) and Bioelectric Branching

4.1 Exact `OEIS A000055` Enumeration via `ExploreFromCanonicalStatesOnly`

In `LimbformDB 2.1.0`, evolving L_1 from a single base joint $[[0]]$ for k steps produces all unlabelled trees on $V = k+1$ vertices. Without canonical deduplication, the k-th step has k available vertices to sprout from, generating $k!$ raw states ($\sum_{k=0}^{11} \max(1, k!) = 43{,}954{,}714$ raw states at $V = 12$). With `ExploreFromCanonicalStatesOnly -> True`, `hg_evolve` expands only one representative per McKay IR class and uses `quotient_multiplicity_core.hpp` to reconstruct the exact factorial raw counts in 85 **ms**, yielding the exact `OEIS A000055` sequence $(1, 1, 1, 2, 3, 6, 11, 23, 47, 106, 235, 551;$ 987 cumulative trees, a 44,533.65× **compression**) and matching 35/35 (100.0%) of `LimbformDB` segment-tree classes with ≤ 12 joints (532/555 = 95.86% of the database; Figure 1D).

4.2 Bioelectric `RuleWeights` Along the Anterior–Posterior Crop Axis

In a 2-step regeneration from an excised trunk $[[0]]$, the anterior blastema applies $D_1 \circ L_1$ (`Head`), while the posterior blastema branches in the multiway `BranchialGraph` between normal $D_2 \circ L_1$ (`Tail`, producing Wild-Type `#1` 1H1T) and depolarized $D_1 \circ L_1$ (`Head`, producing Bipolar Double-Head `#2` 2H0T). Modeling the posterior `RuleWeights` ratio via Boltzmann electrotonic decay along the normalized trunk crop coordinate $p \in \{0, \frac{1}{3}, \frac{2}{3}, 1\}$ ($P_{2\text{H0T}}(p) = [1 + \exp(-\alpha(p - p_{1/2}))]^{-1}$, with $P_{2\text{H0T}} = 0$ on intact-head `tail_crop` controls) fits Oviedo et al. (2010) Series S1 (`1-octanol`, $n = 688$) with $\alpha = 7.45$, $p_{1/2} = 0.565$, **Pearson** $r = 0.9823$ (MAE = 5.45%), and predicts the held-out triple innexin RNAi Series S2 (`Dj-inx-12 + Dj-inx-5/13`, $n = 99$) with **Pearson** $r = 0.9088$ (Figure 1E).

5 Machine-Checked Formal Verification in Lean 4 (`0 sorry`)

All four governing algebraic laws are formalized and verified by the Lean 4 kernel in `morphoeval/lean/loops/Loop` (`0 sorry`, 0 custom axioms):

1. **Theorem 1** (`wolfram_dpo_edge_replacement_tree_euler_invariant`)**:** Proves that under DPO vertex non-deletion (`SPEC.md` §2), every k-step sprouting derivation preserves

$V(k) \geq V_0$, increments $\Delta V = 1$ and $\Delta E = 3$ per step, and maintains the tree Euler invariant $V(k) = L(k) + 1$ for all $k \in \mathbb{N}$.

2. **Theorem 2 (`mckay_ir_zagreb_degree_separates_star_and_chain_isomers`):** Proves that for all $n = k + 4 \geq 4$, the First Zagreb degree-square invariant $Z_1 = \sum d_i^2$ of the Star tree $K_{1,n-1}$ strictly exceeds that of the Path chain P_n by exact gap $(k+1)(k+2) \geq 2$, proving that radial multi-headed stars (`#3`, `#12`) and linear chains (`#119`, `#128`) can never be isomorphic.

3. **Theorem 3 (`quotient_exploration_strictly_compresses_multiway_search`):** Proves that whenever at least one isomorphism class has orbit multiplicity $m^* \geq 2$ among $C \geq 1$ classes, quotient exploration strictly compresses the multiway state space ($C <$ RawStates).

4. **Theorem 4 (`dpo_two_phase_rule_reducibility_conservation`):** Proves region-count conservation $\mathrm{Tr}+H+Ta = V_0+L$ and the exact additive step-cost reduction $k_{\mathrm{prim}} = L+H+Ta$ under the 3-rule primitive basis $\mathcal{B}_3 = \{L_1, D_1, D_2\}$.

6 Discussion: Unifying Lobo's Regulated Graph-Grammar and Target-Morphology Corpus (2010–2015)

A remarkable historical feature of Daniel Lobo's research trajectory is that it spans two seemingly separate mathematical paradigms that had never previously been connected in executable form:

1. **String- and Boolean-Regulated Graph Grammars (Málaga, 2010–2012):** Before creating `PlanformDB` and `LimbformDB` at Tufts, Lobo, Vico, and Dassow developed the formal theory of *graph grammars with string-regulated rewriting* (`ELSA`/`ELSA3`) [3, 4] and *Boolean-network-regulated graph grammars* [5]. In `ELSA`, development starts from a single zygote edge and applies four edge-replacement operations—`Duplicate` (D), `Split` (S), `Resize` (R), and `Connect` (C)—controlled either by an expression-domain substring or by an intracellular Kauffman Boolean network with a differentiation node.

2. **Graph-Encoded Phenotype Databases, Linear Target-Morphology Encodings, and PDE Network Inference (Tufts, 2013–2015):** When Lobo and Levin built `PlanformDB` (2013) [6, 7] and `LimbformDB` (2014) [8, 9], they used graphs to store *static resultant morphologies* and compared them via heuristic *graph edit distance* (GED). However, in their subsequent theoretical and reverse-engineering papers, they did *not* use Lobo's 2011 graph grammars to generate the `PlanformDB` graphs: instead, Lobo et al. (2014, *J. R. Soc. Interface*) [10] contrasted *nonlinear recursive encodings* (such as L-systems, where solving the inverse problem for trophic antler memory or bipolar planaria is intractable because small rule edits trigger global branch cascades) against *linear encodings* of target morphology, while Lobo & Levin (2015, *PLOS Comput. Biol.*) [11] used evolutionary search over continuous reaction–diffusion PDEs on a fixed 2D domain to infer β-catenin/*notum*/*wnt1* pathways.

Our Multiway DPO Hypergraph Rewrite System ($\mathcal{B}_3 = \{L_1, D_1, D_2\}$) provides the missing formal bridge between Lobo's 2011 regulated graph grammars and his 2013–2015 regenerative databases:

- **Connecting `ELSA` Edge-Replacement to `PlanformDB` & `LimbformDB`:** Lobo, Vico & Dassow's (2011) [4] edge-splitting production and differentiation bit map directly onto our DPO edge-replacement basis $\mathcal{B}_3$: the universal budding rule L_1 is the topological dual of `ELSA`'s division rule, while the unary replacements D_1 (`Blastema` $\to$ `Head`) and D_2 (`Blastema` $\to$ `Tail`) formalize the terminal differentiation lock in pure DPO hypergraph semantics (`SPEC.md` §2).

- **Resolving the 2014 "Nonlinear vs. Linear Inverse Problem" Paradox [10]:** In Lobo et al. (2014, Fig. 5) [10], recursive rewriting grammars (L-systems) were classified as *nonlinear encodings* whose inverse problem is intractable because every rule fires globally at

every step, amplifying any small edit across the entire tree. Our Two-Phase Decomposition (Proposition 3.1 and Lean Theorem 4) shows *why* regenerative graph rewriting escapes that paradox: because D_1 and D_2 consume the pluripotent tag [[0]], differentiated regions are **terminally locked** against further L_1 budding, and Phase I topological budding (L_1) commutes with Phase II local fate specification (D_1, D_2). Consequently, changing the bioelectric `RuleWeights` at a single wound site alters only that branch's local fate (D_1 vs. D_2) or adds a single local tine (L_1) at exact additive step cost $k_{\text{prim}} = L + H + Ta$—giving a recursive DPO graph grammar the **local invertibility of a linear encoding**.

- **Replacing Heuristic Graph Edit Distance with Exact 64-Bit McKay IR Hashes:** Where `Planform` (2013) and the 2015 PDE evolutionary search [11] relied on $O(V^3)$ heuristic graph-edit or concentration-grid distances, `HypergraphRewritingEngine`'s McKay IR canonicalizer (`ir_core.hpp`) assigns every morphology an exact, schedule-independent 64-bit isomorphism hash and compresses the 11! multi-step tree search space by 44,534× in 85 ms.

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